AP CALCULUS AB

Unit 1 • Practice • Answer key

Limits and Continuity

AP-style practice • Multiple choice and free response • Answer key with explanations

About this practice set

The questions follow the format of the AP Calculus AB exam and cover exactly lessons 1-1 to 1-9 of the guided notes. Part A and Part B are multiple choice; Part C is free response. **No calculator** on any part. Each question shows the lesson it practices.
Suggested timing: Part A 30 min • Part B 15 min • Part C 45 min.

Part A • Multiple choice • No calculator

Directions • Part A

A calculator may **not** be used. Solve each problem, then choose the best of the four answer choices.

Question 1 (1-2)

NO CALCULATOR

$\lim_{x\to 3} \dfrac{x^2-9}{x^2-2x-3}$ is

(A) 0

(B) $\dfrac{3}{2}$

(C) 3

(D) nonexistent

Solution

Answer: (B). $\dfrac{(x-3)(x+3)}{(x-3)(x+1)} = \dfrac{x+3}{x+1} \to \dfrac{6}{4} = \dfrac{3}{2}$.

Question 2 (1-2)

NO CALCULATOR

If $f(x) = \begin{cases} x^2+1, & x<2 \\ 3x-1, & x\geq 2, \end{cases}$ then $\lim_{x\to 2} f(x)$ is

(A) 5

(B) 4

(C) 6

(D) nonexistent

Solution

Answer: (A). Left: 5. Right: $6-1=5$. Equal, so the limit is 5.

Question 3 (1-3) NO CALCULATOR

$\lim_{x\to\infty} \dfrac{4x^3 - x}{2x^3 + 7x^2}$ is

(A) 0
(B) $\frac{4}{7}$
(C) 2
(D) nonexistent

Solution
Answer: (C). Same degree: ratio of the leading coefficients, $\frac{4}{2} = 2$.

Question 4 (1-3) NO CALCULATOR

Which line or lines are vertical asymptotes of the graph of $h(x) = \dfrac{x+1}{x^2-1}$?

(A) $x = 1$ and $x = -1$
(B) $x = -1$ only
(C) $y = 0$ only
(D) $x = 1$ only

Solution
Answer: (D). $\frac{x+1}{(x-1)(x+1)}$: $x+1$ cancels (hole at $x=-1$); $x-1$ stays (asymptote $x=1$).

Question 5 (1-3) NO CALCULATOR

$\lim_{x\to 2^-} \dfrac{x+1}{x-2}$ is

(A) $-\infty$
(B) $+\infty$
(C) $-\frac{3}{2}$
(D) 0

Solution
Answer: (A). Numerator near 3, denominator small and negative: $-\infty$.

Question 6 (1-4) NO CALCULATOR

For what value of k is $f(x) = \begin{cases} kx+3, & x \le 2 \\ x^2 - k, & x > 2 \end{cases}$ continuous at $x = 2$?

(A) 1
(B) $\frac{1}{3}$
(C) $-\frac{1}{3}$
(D) $\frac{7}{3}$

Solution
Answer: (B). $2k+3 = 4-k$, so $3k = 1$ and $k = \frac{1}{3}$.

Question 7 (1-4) NO CALCULATOR

What type of discontinuity does $g(x) = \dfrac{x^2 - 4}{x - 2}$ have at $x = 2$?

(A) Jump
(B) Infinite
(C) Removable
(D) None

Solution
Answer: (C). $\lim_{x \to 2} g(x) = \lim(x + 2) = 4$ exists, but $g(2)$ is undefined: a hole.

Question 8 (1-5) NO CALCULATOR

The function f is continuous on $[1, 5]$ with $f(1) = -2$ and $f(5) = 6$. Which statement must be true?

(A) $f(3) = 2$
(B) f is increasing on $[1, 5]$.
(C) The maximum of f on $[1, 5]$ is 6.
(D) $f(c) = 0$ for some c in $(1, 5)$.

Solution
Answer: (D). IVT: 0 is between -2 and 6.

Question 9 (1-5) NO CALCULATOR

Which function is guaranteed by the Extreme Value Theorem to have an absolute maximum on the given interval?

(A) $f(x) = x^2 - 3x$ on $[0, 4]$
(B) $f(x) = \dfrac{1}{x}$ on $(0, 1]$
(C) $f(x) = x$ on $(0, 2)$
(D) $f(x) = \dfrac{1}{x - 1}$ on $[0, 2]$

Solution
Answer: (A). Continuous on a closed interval.

Question 10 (1-6) NO CALCULATOR

What is the average rate of change of $f(x) = x^3 - 2x$ on $[-1, 2]$?

(A) 3
(B) 1
(C) $\dfrac{5}{3}$
(D) -1

Solution
Answer: (B). $f(2) = 4$, $f(-1) = 1$: $\frac{4-1}{3} = 1$.

Question 11 (1-6) NO CALCULATOR

$V(t)$ is the volume of water in a tank, in liters, t minutes after noon, with $V(2) = 50$ and $V(8) = 20$. Which is the average rate of change of V on $[2, 8]$?

(A) -30 liters
(B) 5 liters per minute
(C) -5 liters per minute
(D) $-\frac{1}{5}$ minutes per liter

Solution
Answer: (C). $\frac{20-50}{8-2} = -5$ liters per minute.

Question 12 (1-7) NO CALCULATOR

At which point is $f(x) = |x - 2| + 1$ continuous but not differentiable?

(A) $x = 0$
(B) $x = 1$
(C) $x = -2$
(D) $x = 2$

Solution
Answer: (D). Sharp corner at $x = 2$.

Question 13 (1-8) NO CALCULATOR

The function f is differentiable with $f(1) = 4$ and $f(5) = -4$. Which statement must be true?

(A) $f'(c) = -2$ for some c in $(1, 5)$
(B) $f'(c) = 2$ for some c in $(1, 5)$
(C) $f'(c) = -8$ for some c in $(1, 5)$
(D) $f(3) = 0$

Solution
Answer: (A). MVT: $\frac{-4-4}{5-1} = -2$.

Question 14 (1-9) NO CALCULATOR

The depth of a river, $D(t)$, is differentiable. A question asks: "Was the depth ever changing at exactly the average rate for the day?" Which theorem answers it?

(A) Intermediate Value Theorem
(B) Mean Value Theorem
(C) Extreme Value Theorem
(D) None of them

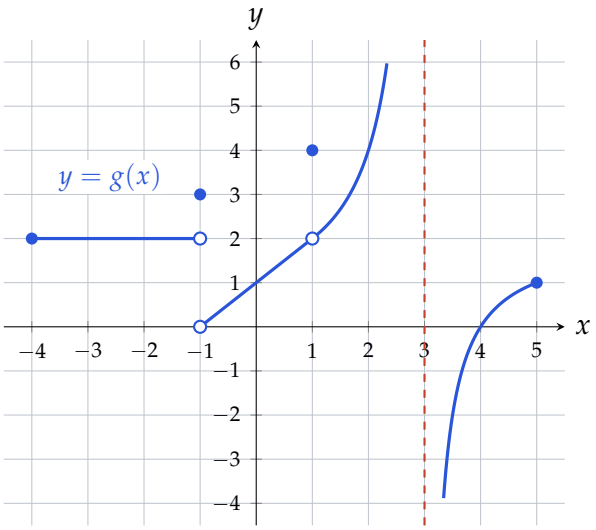
Solution
Answer: (B). A rate of change (slope) equal to an average rate: MVT.

Part B • Multiple choice with a graph • No calculator

Directions • Part B

A calculator may **not** be used. Read the values you need from the graph.

Questions 15 to 20 refer to the graph of the function g below.



The domain of g is $[-4,3) \cup (3,5]$. The line $x = 3$ is a vertical asymptote.

Question 15 (1-1) NO CALCULATOR

Which of the following limits does **not** exist?

(A) $\lim_{x \to 1} g(x)$
(B) $\lim_{x \to -1^+} g(x)$
(C) $\lim_{x \to -1} g(x)$
(D) $\lim_{x \to 4} g(x)$

Solution
Answer: (C). At $x = -1$ the left-hand limit is 2 and the right-hand limit is 0.

Question 16 (1-1) NO CALCULATOR

What is $\lim_{x \to 1} g(x)$?

(A) 4
(B) 2
(C) 3
(D) nonexistent

Solution
Answer: (B). Both pieces approach the open circle $(1,2)$; $g(1) = 4$ does not change the limit.

Question 17 (1-3) NO CALCULATOR

Which statement about the behavior of g near $x = 3$ is true?

(A) $\lim_{x\to 3^-} g(x) = +\infty$ and $\lim_{x\to 3^+} g(x) = -\infty$
(B) $\lim_{x\to 3} g(x) = +\infty$
(C) $\lim_{x\to 3} g(x) = 2$
(D) $g(3) = 2$

Solution
Answer: (A). The graph rises without bound on the left of $x = 3$ and falls without bound on the right.

Question 18 (1-4) NO CALCULATOR

At which value of x does g have a removable discontinuity?

(A) $x = -1$
(B) $x = 3$
(C) $x = -4$
(D) $x = 1$

Solution
Answer: (D). At $x = 1$ the limit is 2 but $g(1) = 4$.

Question 19 (1-6) NO CALCULATOR

What is the average rate of change of g on $[-4, 0]$?

(A) $\frac{1}{4}$
(B) -1
(C) $-\frac{1}{4}$
(D) $\frac{3}{4}$

Solution
Answer: (C). $g(-4) = 2$, $g(0) = 1$: $\frac{1-2}{4} = -\frac{1}{4}$.

Question 20 (1-7) NO CALCULATOR

At which value of x in $(-4, 3)$ is g differentiable?

(A) $x = 0$
(B) $x = -1$
(C) $x = 1$
(D) $x = 3$

Solution
Answer: (A). At $x = 0$ the graph is a straight line (slope 1). g is not continuous at $x = -1$ (jump), $x = 1$ (hole) or $x = 3$ (asymptote), so it is not differentiable there.

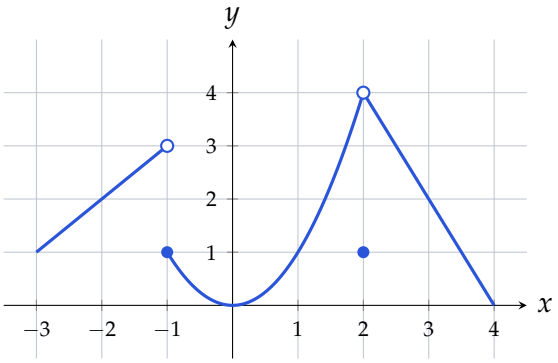
Part C • Free response • No calculator

Directions • Part C

Show all of your work: your answers are graded on the correctness and completeness of your methods as well as on the final answers. Justify answers with mathematical reasons, naming the theorem or definition you use.

Question 21 (Graph analysis) NO CALCULATOR

The graph of the function f on $[-3,4]$ is shown. It consists of a line segment for $-3 \le x < -1$, the parabola $y = x^2$ for $-1 \le x < 2$, and a line segment for $2 < x \le 4$. Also $f(-1) = 1$ and $f(2) = 1$.



(a) Find $\lim_{x\to -1^-} f(x)$ and $\lim_{x\to -1^+} f(x)$. Does $\lim_{x\to -1} f(x)$ exist? Give a reason for your answer.
(b) Find $\lim_{x\to 2} f(x)$. Is f continuous at $x = 2$? Name the type of discontinuity, if any, and explain how $f(2)$ could be redefined to make f continuous at $x = 2$.
(c) Is f differentiable at $x = -1$? At $x = 0$? Explain each answer.
(d) $f(1) = 1$ and $f(3) = 2$. Explain why the Intermediate Value Theorem cannot be used to guarantee a value c in $[1,3]$ with $f(c) = 1.5$. Does such a value exist? Justify.

Solution

(a) $\lim_{x\to -1^-} f(x) = 3$ and $\lim_{x\to -1^+} f(x) = (-1)^2 = 1$. The one-sided limits are not equal, so $\lim_{x\to -1} f(x)$ does not exist.
(b) $\lim_{x\to 2^-} f(x) = 2^2 = 4$ and $\lim_{x\to 2^+} f(x) = 8 - 2(2) = 4$, so $\lim_{x\to 2} f(x) = 4$. But $f(2) = 1 \neq 4$, so f is not continuous at $x = 2$. This is a removable discontinuity; redefining $f(2) = 4$ makes f continuous at $x = 2$.
(c) At $x = -1$: no, f is not continuous there (jump), and a function is not differentiable where it is not continuous. At $x = 0$: yes, the graph is the smooth parabola $y = x^2$ there, with a horizontal tangent line (slope 0).
(d) The IVT requires f to be continuous on $[1,3]$, but f is not continuous at $x = 2$. Such a value does exist: on $[1,2)$, $f(x) = x^2 = 1.5$ when $c = \sqrt{1.5} \approx 1.22$, which is in $[1,3]$.

Question 22 (Algebraic limits and continuity) NO CALCULATOR

Let $g(x) = \begin{cases} \dfrac{x^2+x-6}{x-2}, & x < 2 \\ cx+1, & x \ge 2 \end{cases}$ where c is a constant, and let $h(x) = \dfrac{x^2+x-6}{x^2-4}$.

(a) Find $\lim_{x\to 2^-} g(x)$. Show the work that leads to your answer.
(b) Find the value of c for which g is continuous at $x = 2$. Justify your answer using the definition of continuity.
(c) Find all values of x where h is not continuous, and classify each discontinuity. Justify with limits.
(d) Find $\lim_{x\to\infty} h(x)$ and $\lim_{x\to -2^+} h(x)$.

Solution

(a) For $x < 2$: $\dfrac{(x+3)(x-2)}{x-2} = x+3$, so $\lim_{x\to 2^-} g(x) = 5$.

(b) $g(2) = 2c+1$ and $\lim_{x\to 2^+} g(x) = 2c+1$. For continuity we need $\lim_{x\to 2^-} g(x) = \lim_{x\to 2^+} g(x) = g(2)$: $2c+1 = 5$, so $c = 2$. Then $g(2) = 5$ and $\lim_{x\to 2} g(x) = 5 = g(2)$.

(c) $h(x) = \dfrac{(x+3)(x-2)}{(x+2)(x-2)}$ is not defined at $x = 2$ and $x = -2$. At $x = 2$: $\lim_{x\to 2} h(x) = \lim_{x\to 2} \dfrac{x+3}{x+2} = \dfrac{5}{4}$ exists: removable discontinuity. At $x = -2$: the numerator of $\dfrac{x+3}{x+2}$ approaches 1 and the denominator approaches 0, so $\lim_{x\to -2^-} h(x) = -\infty$ and $\lim_{x\to -2^+} h(x) = +\infty$: infinite discontinuity (vertical asymptote $x = -2$).

(d) $\lim_{x\to\infty} h(x) = 1$ (equal degrees, leading coefficients 1 and 1). $\lim_{x\to -2^+} h(x) = +\infty$.

Question 23 (Table and theorems) NO CALCULATOR

The temperature W, in degrees Celsius, of a greenhouse is a differentiable function of time t, in hours, for $0 \le t \le 10$. Selected values are given in the table.

t (hours)	0	3	5	8	10
$W(t)$ (°C)	20	24	31	27	22

(a) Must there be a time t in $[0,10]$ at which $W(t) = 30$? Justify your answer.
(b) Explain why there must be at least two times in $[0,10]$ at which $W(t) = 25$.
(c) Find the average rate of change of W on $3 \le t \le 8$. Give units and interpret it.
(d) Must there be a time t in $(5,8)$ at which $W'(t) = -\frac{4}{3}$? Justify. Must W have an absolute maximum on $[0,10]$? Justify.

Solution

(a) Yes. W is differentiable, so it is continuous on $[3,5]$; $W(3) = 24$ and $W(5) = 31$. Since $24 < 30 < 31$, the IVT guarantees a t in $(3,5)$ with $W(t) = 30$.

(b) On $[3,5]$: $24 < 25 < 31$, so $W(t) = 25$ for some t in $(3,5)$. On $[8,10]$: $22 < 25 < 27$, so $W(t) = 25$ for some t in $(8,10)$. These intervals do not overlap: at least two times (IVT twice).

(c) $\dfrac{W(8)-W(3)}{8-3} = \dfrac{27-24}{5} = 0.6$ °C per hour: between hours 3 and 8 the temperature rose on average 0.6 degrees per hour.

(d) Yes: W is differentiable on $(5,8)$ and continuous on $[5,8]$, and $\dfrac{27-31}{8-5} = -\dfrac{4}{3}$, so the MVT applies. Yes: W is continuous on the closed interval $[0,10]$, so the EVT guarantees an absolute maximum (at least 31).